

AWS Certified AI Practitioner

AIF-C01 — Complete Study Guide

Organized by Exam Domain • Cross-Referenced with Official Exam Guide

All 5 Domains • In-Scope Services Highlighted • Practice Q&A Included

Domain	Topic	Weight	~Questions
Domain 1	Fundamentals of AI and ML	20%	~10 Qs
Domain 2	Fundamentals of Generative AI	24%	~12 Qs
Domain 3	Applications of Foundation Models	28% ★ Most	~14 Qs
Domain 4	Guidelines for Responsible AI	14%	~7 Qs
Domain 5	Security, Compliance & Governance for AI	14%	~7 Qs

Exam Detail	Description
Total Questions	65 (50 scored + 15 unscored)
Passing Score	700 / 1,000 (scaled)
Time	90 minutes
Question Types	Multiple choice, Multiple response, Ordering, Matching
NOT on Exam	Building/coding models, feature engineering, hyperparameter math, deploying pipelines
Target Candidate	Up to 6 months exposure to AI/ML on AWS — user, not builder

■ **EXAM TIP:** Domain 3 (Applications of Foundation Models) = 28% — spend the MOST study time here. RAG, prompt engineering, fine-tuning, and evaluation metrics are heavily tested.

Task 1.1 — Basic AI Concepts & Terminologies

The AI Hierarchy (Largest to Smallest Scope)

Category	Definition
Artificial Intelligence (AI)	Broadest field — systems performing tasks that require human-like intelligence (perception, reasoning, decision-making).
Machine Learning (ML)	Subset of AI — systems learn from data to improve performance without explicit programming.
Deep Learning (DL)	Subset of ML — multi-layer artificial neural networks inspired by the human brain. Powers computer vision & NLP.
Generative AI	Subset of DL — creates NEW content (text, images, audio, code, video) from learned patterns. Does NOT require retraining per new task.

■ **EXAM TIP:** Key hierarchy: GenAI $\subset$ Deep Learning $\subset$ ML $\subset$ AI. Generative AI can adapt foundation models WITHOUT retraining for each new task.

Core Terminology

Term	Definition
Algorithm	Set of rules/instructions the model uses to learn from data.
Model	Output of training an algorithm on data — used to make predictions or generate content.
Training	Process of feeding data to an algorithm so it learns patterns and relationships.
Inferencing	Using a trained model to make predictions on new, unseen data.
Tokens	Basic units of text the LLM processes (words, sub-words, characters). Standardizes input data.
Embeddings / Vectors	Numerical vector representations of tokens capturing semantic meaning. Similar concepts = similar vectors.
Bias (model)	Model misses key features — high gap between predicted and actual values → underfitting.
Variance (model)	Model is overly sensitive to training noise → overfitting on training data.
Fairness	AI promotes inclusion, avoids discrimination, upholds responsible values.
Fit	Underfitting (too simple/high bias) Balanced (ideal) Overfitting (too complex/high variance).
LLM	Large Language Model — transformer-based deep learning foundation model trained on massive text datasets.
Hallucination	Model generates plausible-sounding but factually incorrect output.

Chunking	Breaking large documents into smaller pieces for embedding and retrieval (used in RAG).
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Types of Data in AI Models

Data Type	Description & Use
Labeled	Has associated target/label. Used in supervised learning.
Unlabeled	No labels — used in unsupervised learning and FM pre-training at scale.
Tabular	Structured data in rows/columns (CSV, databases, spreadsheets).
Time-series	Values measured at successive time points (stock prices, sensor readings).
Image	Pixel arrays — used in computer vision tasks.
Text	Sequences of characters/tokens — NLP tasks.
Structured	Organized in predefined format (tables, databases). Works with traditional ML.
Unstructured	No fixed format — images, audio, video, raw text. Requires deep learning.

Learning Paradigms

Paradigm	Description
Supervised Learning	Trained on labeled data. Learns input→output mapping. Tasks: classification, regression.
Unsupervised Learning	Trained on unlabeled data. Discovers hidden patterns. Tasks: clustering, dimensionality reduction.
Reinforcement Learning	Agent learns via trial-and-error; earns rewards for correct actions, penalties for wrong. Example: AWS DeepRacer.

Inferencing Types

Type	Description
Batch Inferencing	Offline processing of large datasets all at once. Cost-effective. NOT latency-sensitive. Example: nightly report generation.
Real-time Inferencing	Low latency, persistent endpoint, high throughput. Immediate decisions. Example: chatbots, fraud detection, self-driving cars.

Task 1.2 — Practical Use Cases for AI

When AI/ML IS Appropriate

- Coding explicit rules is **too complex** (spam filtering, fraud detection — too many overlapping variables).
- Scale is challenging — processing **millions** of items that humans cannot review.
- Content creation, personalization, automation, efficiency, and innovation tasks.
- Assisting human decision-making, solution scalability, pattern recognition, anomaly detection.

When AI/ML is NOT Appropriate

- High accuracy/reliability required AND errors are **very costly** (no human oversight).
- Explainability and transparency are essential requirements.
- Problem is ill-defined or constantly changing.
- Cost of implementation outweighs benefits.
- Insufficient quality data.
- Ethical considerations prohibit AI use.
- A **specific deterministic outcome** is needed instead of a prediction.
- Making critical medical diagnoses **without human oversight**.

ML Technique Selection

Technique	Use Case & Example
Classification	Predicts a category label . Binary (fraud/not-fraud) or multi-class (image types). Fraud detection, diagnostics, spam filtering.
Regression	Predicts a continuous value . Price prediction, weather forecasting, life expectancy, demand forecasting.
Clustering	Groups similar data without labels . Customer segmentation, target marketing, recommender systems.
Anomaly Detection	Identifies data points that deviate from the norm. Fraud, network intrusion, equipment failure.
Computer Vision	Interprets images/video. Autonomous driving, medical imaging, facial recognition (Amazon Rekognition).
NLP	Processes human language. Sentiment analysis, translation, chatbots (Amazon Comprehend, Lex, Translate).
Recommendation Systems	Personalizes content/products. Collaborative filtering (Amazon Personalize).
Speech Recognition	Speech to text. Transcription, voice commands (Amazon Transcribe).
Forecasting	Predicts future values from time-series data. Demand planning, inventory.

IN-SCOPE AWS AI/ML Managed Services ★ (Must Know)

Service	Key Capability
Amazon SageMaker AI ★	Full ML platform: build, train, deploy, monitor models. End-to-end ML lifecycle.
Amazon Bedrock ★	Access foundation models from multiple providers via unified API. Serverless. Supports RAG, agents, guardrails.
Amazon Comprehend ★	NLP: extract key phrases, entities, sentiment, PII from text. Language detection.
Amazon Transcribe ★	Automatic speech recognition (ASR). Speech-to-text. Real-time or batch.
Amazon Translate ★	Neural machine translation. 75+ languages.

Amazon Polly ★	Text-to-speech. Lifelike voices. Powers IVR systems.
Amazon Lex ★	Build conversational AI (chatbots, IVR). Combines ASR + NLU. Powers Alexa.
Amazon Rekognition ★	Image/video analysis. Object detection, facial recognition, content moderation.
Amazon Textract ★	Extract text, handwriting, tables, forms from scanned documents (beyond basic OCR).
Amazon Kendra ★	Intelligent ML-powered document search. Enterprise knowledge base.
Amazon Personalize ★	Real-time personalized product/content recommendations.
Amazon Fraud Detector ★	Detect online fraud, account takeovers, payment fraud.
Amazon Q Business ★	Enterprise AI assistant. Answers from company repos, code, enterprise systems.
Amazon Q Developer ★	AI coding assistant. Code generation, completion, debugging.
Amazon Nova ★	Amazon's family of foundation models available through Bedrock.
Amazon A2I ★	Augmented AI — human review loop for ML predictions. Route low-confidence predictions to humans.

Task 1.3 — ML Development Lifecycle

ML Pipeline Stages

Stage	Description
1. Business Goal	Define value, budget, KPIs, success criteria. ML starts with a business objective.
2. ML Problem Framing	Translate business problem into ML problem. Assess data/skill feasibility.
3. Data Collection	Gather raw data into centrally accessible place.
4. EDA / Pre-processing	Visualize and transform raw data. Remove noise, normalize, handle missing values.
5. Feature Engineering	Create, transform, extract, select variables for the model. Critical for performance.
6. Model Training	Feed training data to algorithm. Produce trained model.
7. Hyperparameter Tuning	Optimize model parameters (NOT the same as training weights).
8. Evaluation	Measure performance on validation and test sets. Check for over/underfitting.
9. Deployment	Integrate trained model into production environment via API.
10. Monitoring	Track model quality, data drift, and bias over time. Retrain when performance degrades.

SageMaker AI Services by Pipeline Stage

SageMaker Service	Purpose
SageMaker Data Wrangler	Low-code/no-code data prep: import, transform, featurize. Visual interface.
SageMaker Feature Store	Create, share, manage ML features. Reusable across projects.

SageMaker Training Jobs	Managed ML training with built-in algorithms or custom code.
SageMaker Experiments	Track combinations of data, algorithms, parameters. Hyperparameter tuning.
SageMaker Automatic Tuning	Automated hyperparameter optimization (runs many jobs, picks best).
SageMaker Model Monitor	Monitor production models for quality, bias, data drift. Set alerts. ★
SageMaker Clarify	Detect bias in datasets and models. Feature importance scores. ★
SageMaker Model Cards	Document model details: intended use, risk rating, training info. Governance. ★
SageMaker Canvas	No-code ML — generate predictions without writing code.
SageMaker JumpStart	Deploy/fine-tune 150+ pre-trained open-source models with one click.
SageMaker Studio	Web-based unified IDE for all SageMaker tasks.
SageMaker Ground Truth	Data labeling + human-in-the-loop (RLHF data annotation).

Model Deployment Options

Type	When to Use
Real-time	Persistent endpoint. Low latency, high throughput. Chatbots, fraud detection. Best for interactive/immediate needs. ★
Batch Transform	Offline processing. Large datasets. No persistent endpoint. Cost-effective for infrequent inference.
Asynchronous	Queued requests. Payloads up to 1 GB. Processing up to 1 hour. Near real-time SLA.
Serverless	No infrastructure management. Scales to zero. Best for sporadic/idle traffic with cold-start tolerance.

■ **EXAM TIP:** Latency order (lowest to highest): Real-time → Asynchronous → Batch Transform. Serverless has cold-start latency but zero idle cost.

Model Performance Metrics

Metric	What It Measures
Accuracy	Correct predictions ÷ total. Limited when class imbalance exists.
Precision	$TP \div (TP+FP)$. When false positives are costly. 'Of all predicted positives, how many were correct?'
Recall	$TP \div (TP+FN)$. When false negatives are costly. 'Of all actual positives, how many did we catch?'
F1 Score	Harmonic mean of Precision and Recall. Best for imbalanced classes. ★
AUC-ROC	Area Under the Curve of true positive vs. false positive rates across thresholds. Higher = better separation.
MSE	Mean Squared Error — regression. Average squared error. Smaller = better.
R ²	Goodness-of-fit for linear regression. Closer to 1.0 = model explains more variance.

■ **EXAM TIP:** Overfitting = High Variance (good on train, bad on test). Underfitting = High Bias (bad on both). Goal = Low Bias + Low Variance.

MLOps Key Concepts

- **MLOps** = people + process + technology to operationalize the ML lifecycle end-to-end (DevOps for ML).
- **Key principles:** Version control, Automation (CI/CD), Continuous training, Continuous monitoring, Repeatability.
- **Benefits:** Productivity, Reliability, Repeatability, Auditability, Data/model quality enforcement.
- **Continuous Integration** — extends testing to data and models in the pipeline.
- **Continuous Delivery** — automatically deploys the newly trained model.
- **Continuous Training** — automatically retrains ML models for redeployment.
- **Continuous Monitoring** — monitors data quality and model performance metrics.

Task 2.1 — Basic Concepts of GenAI

Foundation Models (FMs)

- Trained on **internet-scale unlabeled data** (self-supervised learning — no manual labels required).
- One FM can be adapted to **multiple tasks** without retraining: text generation, summarization, extraction, image generation, chatbots, Q&A, code generation.
- Quality depends on both the training data AND the ML algorithm used.
- FMs serve as the starting point for building specialized, domain-specific models.
- Key difference from traditional ML: traditional models do **one task**; FMs do **many tasks**.

FM Lifecycle

Phase	Description
Data Selection	Large-scale unlabeled data from diverse sources. Easier to obtain than labeled data.
Pre-Training	Self-supervised learning. Model learns meaning, context, and word relationships from raw text.
Continuous Pre-Training	Extend pre-training with new data to expand the model's knowledge without full retraining.
Optimization	Prompt engineering, RAG, or fine-tuning to improve task-specific performance.
Evaluation	Assess with metrics (ROUGE, BLEU, BERTScore) and human evaluation. Check business fit.
Deployment	Integrate model into apps via APIs or software systems.
Feedback & Improvement	Monitor, collect user feedback, retrain/fine-tune iteratively to improve.

Key GenAI Concepts

Concept	Explanation
Tokens	Basic units of text processed by LLMs (words, sub-words, characters). Cost is token-based.
Embeddings/Vectors	Numerical representations of tokens. Similar meaning = similar vectors. Used in RAG & semantic search.
Chunking	Breaking documents into smaller pieces for embedding & retrieval in vector databases.
Prompt Engineering	Designing inputs to guide FM output — fastest/cheapest optimization approach.
Context Window	Maximum tokens that fit in one prompt. Limits how much text the model can 'see' at once.

GenAI Model Architectures

Architecture	Description
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Transformer-based LLMs	Most common. Bidirectional/unidirectional. Examples: GPT, BERT, Claude, Llama. Text understanding & generation.
Diffusion Models	Start with pure noise; forward diffusion adds noise; reverse diffusion denoises to produce output. Primarily for image generation.
Multimodal Models	Process/generate multiple modalities simultaneously (text + image + audio). Video captioning, text-to-image.
GANs	Generator + Discriminator compete. Generator creates synthetic data; Discriminator detects fakes. Image synthesis.
VAEs	Encoder (maps to latent space) + Decoder (generates from samples). Structured probabilistic generation.

Task 2.2 — Capabilities & Limitations of GenAI

Advantages of Generative AI

Advantage	Description
Adaptability	Adapts to diverse tasks and domains by learning from data.
Responsiveness	Generates content in real-time — enables interactive chatbots and virtual assistants.
Simplicity	Simplifies content creation tasks through automation.
Creativity & Exploration	Generates novel ideas, designs, and solutions by combining patterns in new ways.
Data Efficiency	Some models learn from relatively small amounts of data.
Personalization	Creates tailored content for individual preferences.
Scalability	Produces large volumes of content quickly once trained.

Disadvantages / Risks of GenAI

Disadvantage	Description & Mitigation
Hallucinations	Model generates plausible but factually incorrect outputs. Mitigation: RAG, user verification, guardrails.
Nondeterminism	Same input → different outputs. Mitigation: test multiple times; set temperature parameters.
Interpretability	Difficult to understand why the model produced a specific output.
Inaccuracy	Outputs may be incorrect even when presented confidently.
Toxicity	Can generate offensive/inappropriate content. Mitigation: guardrails, data curation.
Intellectual Property	May reproduce copyrighted training data verbatim. Legal risk.
Data Security/Privacy	Sharing sensitive data with a model may violate privacy laws. Mitigation: encryption, anonymization.
Regulatory Violations	Outputs may inadvertently expose PII or violate regulations.

Factors for Selecting a GenAI Model

- **Model type** — text, image, audio, multimodal; different providers excel at different tasks.
- **Performance requirements** — accuracy, reliability. Test against diverse domain-specific datasets.
- **Constraints** — compute (GPU/CPU/memory), data availability, deployment location (cloud vs. on-prem).
- **Capabilities** — specific modality support, context window size, multi-language support.
- **Compliance** — bias risk, fairness, transparency, accountability, hallucination rate.
- **Cost** — larger models = more precise but expensive; smaller = cheaper and faster.
- **Latency** — real-time apps need low-latency models with fast inference.

Business Metrics for GenAI Applications

Metric	What It Measures
User Satisfaction	Customer feedback, NPS. Measures trust and engagement with AI outputs.
ARPU (Avg Revenue/User)	Revenue per user attributable to GenAI. Measures monetization effectiveness.
Cross-Domain Performance	Model's ability to perform consistently across different domains.
Conversion Rate	% users completing desired actions (purchases, sign-ups) driven by AI recommendations.
Efficiency	Resource utilization improvements, computation time reduction, scalability.
ROI	Cost savings + time savings + quality improvements vs. total AI investment.
Customer Lifetime Value	Long-term value of customers engaging with AI-powered experiences.

Task 2.3 — AWS Infrastructure for GenAI

Core AWS GenAI Services (All IN-SCOPE ★)

Service	Description
Amazon Bedrock ★	Fully managed. Unified API for FMs from Amazon, Anthropic, AI21, Cohere, Meta, Mistral, Stability AI. Serverless. Supports RAG (Knowledge Bases), agents, guardrails, model evaluation.
Amazon SageMaker JumpStart ★	One-click deployment & fine-tuning of 150+ pre-trained open-source models.
Bedrock Knowledge Bases ★	Managed RAG — store embeddings, connect knowledge to FMs. Reduces hallucinations.
Bedrock Agents ★	Automate multi-step tasks; orchestrate between FM and backend systems.
Bedrock Guardrails ★	Block topics, filter harmful content, redact PII, enforce responsible AI policies.
Bedrock Model Evaluation ★	Compare & select FMs. Automatic metrics (accuracy, robustness, toxicity) and human evaluation.
Amazon Q Business ★	Enterprise AI assistant — fast answers from company data, repos, systems.
Amazon Q Developer ★	AI-powered coding assistant. Code generation, completion, security scanning.

Bedrock Data Automation	Automates data extraction and transformation workflows using GenAI.
PartyRock (Bedrock Playground)	No-code FM experimentation and prototyping platform.

Available FM Providers on Amazon Bedrock

Provider	Models & Strengths
Amazon	Nova (text, image, multimodal). Titan (text, embeddings, image). General-purpose, enterprise-ready.
Anthropic	Claude (Haiku = fast/cheap, Sonnet = balanced, Opus = most capable). Content, code, reasoning, analysis.
AI21 Labs	Jurassic / Jamba. Long-form text, summarization, paraphrasing.
Cohere	Command (text generation, extraction, Q&A). Embed (semantic search). Enterprise reliability.
Meta	Llama models. Q&A, chat, summarization, sentiment analysis. Open-source weights.
Mistral AI	Efficient text generation. European-origin models. Multiple sizes.
Stability AI	Stable Diffusion. Photorealistic image generation from text.

AWS GenAI Service Cost Considerations

- **Token-based pricing** — pay per input + output token (on-demand). More tokens = higher cost.
- **Provisioned throughput** — reserve capacity in advance. Predictable but higher cost.
- **Custom models** — fine-tuning and deploying custom FMs incurs additional training costs.
- **Responsiveness/availability** — lower latency + higher availability = higher cost.
- **Redundancy/multi-region** — cross-AZ/cross-region deployments add cost.
- **GPU instances** — high-performance compute for GenAI costs more than CPU.

■ **EXAM TIP:** This domain = 28% of the exam — roughly 14 questions. Master RAG, prompt engineering, fine-tuning, and evaluation metrics thoroughly.

Task 3.1 — Design Considerations for FM Applications

Pre-Trained Model Selection Criteria

Criterion	Consideration
Cost	Licensing fees + inference compute + fine-tuning costs. Weigh against expected benefits.
Modality	Text, image, audio, video, or multimodal. Match to your application's needs.
Latency	Real-time = low latency. Evaluate inference speed vs. available compute resources.
Multi-lingual	Verify model supports all required languages for your use case.
Model size	Larger = higher accuracy but higher compute. Balance with resources and budget.
Model complexity	Complex models handle advanced tasks but are harder to deploy and optimize.
Customization	Can the model be fine-tuned on your domain? Requires additional data and compute.
Input/output length	Context window limits. Long-form apps need large context windows.
Prompt caching	Cache common prompts to reduce cost and latency for repeated requests.
Responsibility	Evaluate bias risk, misinformation potential, societal impact, licensing restrictions.

Inference Parameters and Their Effects

Parameter	Effect on Output
Temperature	Controls randomness. Low (0) = deterministic/focused outputs. High (1+) = creative/varied. Critical for managing hallucination risk. ★
Top-P (nucleus)	Model considers only tokens making up top P% probability mass. Low = focused; high = diverse.
Input/output length	Max tokens to process or generate. Affects cost (token pricing) and context window usage.
Stop sequences	Special tokens that signal the model to stop generating further output.

Retrieval Augmented Generation (RAG) ★

- **RAG** = combines a retrieval system (knowledge base) with a generative LLM to produce grounded, accurate responses.
- **Retrieval system** — fetches relevant documents from knowledge base, web, or data sources using similarity search.
- **Generative model** — synthesizes retrieved info with its own knowledge to produce a coherent response.
- **RAG does NOT modify model weights** (unlike fine-tuning) — retrieves context at inference time.

- **Benefits:** Reduces hallucinations, improves accuracy, incorporates proprietary/recent data, no retraining needed.
- **AWS Service:** Amazon Bedrock Knowledge Bases manages the RAG pipeline end-to-end.

RAG Business Applications

- Customer service chatbots — answer product/policy questions from a product knowledge base.
- Legal research assistant — retrieve laws, regulations, case precedents.
- Healthcare Q&A — answer complex medical queries from clinical guidelines.
- Enterprise document search — find documents scattered across multiple repositories.

Vector Databases for Embedding Storage (IN-SCOPE ★)

AWS Vector DB Service	Use Case
Amazon OpenSearch Service ★	Primary choice. Provisioned or Serverless. Supports k-NN and cosine similarity search at scale.
Amazon Aurora (pgvector) ★	PostgreSQL-compatible. pgvector extension for embedding storage alongside relational data.
Amazon RDS for PostgreSQL ★	Relational DB with pgvector extension for vector search.
Amazon Neptune ★	Graph database with vector search support. Graph-based RAG.
Amazon DocumentDB ★	MongoDB-compatible document DB with vector search support.

■ **EXAM TIP:** Amazon OpenSearch Service is the most commonly tested vector database. Know it supports k-NN (k-nearest neighbors) and cosine similarity.

FM Customization Cost-Accuracy Trade-offs ★

Approach	Description	Cost	Accuracy
1. Use existing FM as-is	Zero effort. Use pre-trained model directly for general tasks.	\$ Lowest	Lowest
2. Prompt Engineering	Craft optimized prompts/templates. No training needed. Fastest.	\$ Very Low	Moderate
3. RAG	Retrieve external context at inference time. No weight changes.	\$\$ Low-Med	High (factual/current)
4. Fine-tuning	Additional training on task-specific labeled data. Changes weights.	\$\$\$ Med-High	High (specialized tasks)
5. Continuous Pre-training	Further pre-train FM on domain corpus. Expands knowledge.	\$\$\$\$ High	High (domain adaptation)
6. Build from scratch	Train entirely new FM architecture. Research or unique needs.	\$\$\$\$\$ Highest	Highest (full control)

■ **EXAM TIP:** Exam ordering (least to most overhead): Use existing LLM → Prompt Eng. → RAG → Fine-tune → Continuous Pre-train → Build from scratch.

Agents for Amazon Bedrock ★

- **Agents** — orchestrate multi-step tasks between the LLM and backend systems (APIs, databases, services).
- **Task coordination** — manage subtask order and dependencies. Ensure prerequisites are met before proceeding.
- **Actions launch** — run tasks like API calls, DB queries, calculations, file operations.
- **Feedback integration** — collect outcome data to refine the AI model over time.
- **Scalability** — handle multiple instances of multi-step tasks concurrently.
- **Model Context Protocol (MCP)** — standardized protocol for agents to interface with external tools and data sources.
- **Agentic AI** — AI that autonomously plans, decides, and acts to complete goals with minimal human intervention.

Task 3.2 — Effective Prompt Engineering Techniques

Prompt Engineering Fundamentals

- **Prompt** — instructions/input given to an FM that shapes its output.
- **Benefits:** Get best results from model capabilities; bolster safety; provide domain knowledge without fine-tuning; higher quality outputs.
- **Best practices:** Be clear and concise; include relevant context; use specific directives; state outcome requirements; provide example responses.

Prompt Components & Elements

Component/Element	Purpose
Task	What you want the model to DO (the main instruction).
Context	Background information to guide the model's understanding.
Input Data	The specific content to process or respond to.
Output Format	Specify format, length, tone, structure of the expected response.
Negative Prompts	What NOT to include — steer away from unwanted content.
Role	Give the model a persona: 'You are an expert in...'
Examples	Provide sample inputs/outputs to guide the model (few-shot).
Constraints	Boundaries on what the model should/should not do.

Prompting Techniques ★

Technique	Description & Best Use
Zero-shot	No examples given. Model uses general knowledge. Best for large, capable models on simple tasks.
Single-shot	One example provided in the prompt. Clarifies the expected output format.
Few-shot ★	2–5 examples in the prompt. Teaches the model desired style/format. LEAST operational effort for style matching.

Chain-of-Thought (CoT) ★	'Think step-by-step' prompt. Encourages logical reasoning. Best for math, logic, multi-step problems.
Tree-of-Thoughts (ToT)	Explores multiple reasoning paths simultaneously; evaluates intermediate steps. Complex multi-path problems.
RAG (as technique)	Retrieves external knowledge as context for the prompt. Reduces hallucinations. Incorporates fresh data.
Self-consistency	Generate multiple reasoning paths; pick the most consistent answer. Improves accuracy on reasoning tasks.
ReAct prompting	Combines reasoning AND acting — model uses external tools during the reasoning process.
Prompt Templates	Reusable prompt structures with variables for consistent, scalable prompting.
Prompt Routing	Direct different types of prompts to different models or handlers based on content.

■ **EXAM TIP:** Few-shot prompting = provide examples of desired output in the prompt. LEAST operational effort for style/format matching. No training needed.

Prompt Engineering Risks & Limitations

Risk	Description
Prompt Injection	Malicious input overrides original system instructions. User hijacks model behavior.
Prompt Poisoning	Adversarial inputs in training data corrupt model responses for certain triggers.
Prompt Hijacking	Attacker redirects the model to perform unintended actions.
Jailbreaking	User crafts clever prompts to bypass safety guardrails and restrictions.
Exposure	Sensitive system prompts or proprietary data revealed through model outputs.

■ **EXAM TIP:** Guardrails for Amazon Bedrock is the primary AWS tool to mitigate prompt injection, jailbreaking, and harmful content generation.

Task 3.3 — Training & Fine-Tuning Foundation Models

Fine-Tuning Methods ★

Fine-Tuning Method	Description
Instruction Fine-Tuning ★	Train on (instruction → desired response) examples. Improves command-following. Prompt tuning is a type of instruction fine-tuning.
RLHF ★	Reinforcement Learning from Human Feedback. Humans rate outputs; reward model trained from ratings; RL refines the LLM toward human preferences.
Domain Adaptation	Fine-tune on domain-specific corpus (legal docs, medical records). Improves accuracy for that domain. Uses limited domain-specific data.

Transfer Learning	Re-use model trained on Task A as starting point for Task B. Efficient — less additional training needed.
Continuous Pre-Training	Extend pre-training with new/emerging data to keep model current without full retraining.
Distillation	Train a smaller 'student' model to mimic a larger 'teacher' model. Reduces cost/latency while maintaining performance.

■ **EXAM TIP:** Fine-tuning CHANGES model weights. RAG does NOT change model weights — it retrieves at inference time. This distinction is heavily tested.

Data Preparation for Fine-Tuning

- **Quality over quantity:** Well-curated small datasets can significantly improve fine-tuned performance.
- **Specificity:** Fine-tuning data must be highly relevant to the specific target task or domain.
- **Data curation:** Rigorous selection — every data point should contribute to target task learning.
- **Labeling:** Accurate and relevant labels are critical for fine-tuning success (unlike pre-training).
- **Representativeness:** Avoid introducing new biases; ensure data covers the target distribution.
- **Bias checking:** Fine-tuning datasets must not introduce or perpetuate unfair biases.
- **RLHF feedback integration:** Comparison and ranking data serves as the reward function.
- **Pre-training data:** Maximum quantity + diversity from varied public sources. Thorough cleaning.

Task 3.4 — Evaluating Foundation Model Performance

Evaluation Methods

Method	Description
Human Evaluation ★	Humans assess coherence, factuality, and quality. Gold standard but expensive & slow. Best for qualitative assessment.
Benchmark Datasets ★	Curated standardized datasets (GLUE, SuperGLUE, SQuAD, WMT). Objective, scalable, consistent. Best for technical comparison.
Automated Metrics	BLEU, ROUGE, BERTScore, perplexity, F1. Fast and scalable but may not align perfectly with human judgment.
Combined Approach ★	Use BOTH benchmark datasets AND human evaluation. Recommended for production deployment decisions.
Amazon Bedrock Model Evaluation ★	Automatic evaluation (accuracy, robustness, toxicity) + human evaluation (brand alignment, style). Compare FMs side-by-side.

Key FM Evaluation Metrics ★★

Metric	What It Measures & Primary Use Case
ROUGE ★★	Recall-Oriented Understudy for Gisting Evaluation. Measures RECALL of key information. Counts overlapping n-grams between generated and reference text. Primary metric for summarization . Correlates well with human judgment for recall.

BLEU ★★	Bilingual Evaluation Understudy. Measures PRECISION of n-grams. Applies brevity penalty for short translations. Primary metric for machine translation . Unlike ROUGE (recall), BLEU is a precision metric.
BERTScore ★	Uses pre-trained BERT embeddings. Computes cosine similarity between contextual embeddings. Evaluates SEMANTIC SIMILARITY — understands synonyms & paraphrases. Does not rely on exact word matches. More nuanced than BLEU/ROUGE.
Perplexity	Measures how well the model predicts the next token. Lower = better language model performance.
F1 Score	Harmonic mean of precision and recall. Used for classification and entity recognition tasks.

■ **EXAM TIP:** ROUGE = RECALL (summarization). BLEU = PRECISION (translation). BERTScore = SEMANTIC SIMILARITY (meaning). These are the three most tested evaluation metrics.

Benchmark Dataset	What It Evaluates
GLUE / SuperGLUE	Text classification, Q&A, natural language inference. SuperGLUE = harder, compositional.
SQuAD	Stanford Question Answering Dataset. Evaluates question-answering capabilities.
WMT	Workshop on Machine Translation. Series of datasets for evaluating machine translation.

Task 4.1 — Developing Responsible AI Systems

Core Dimensions of Responsible AI ★

Dimension	Definition
Fairness	AI promotes inclusion, prevents discrimination, upholds responsible values and legal norms, builds societal trust.
Explainability	Ability to clearly explain or justify internal model decisions in human-understandable terms.
Privacy & Security	Data is protected from theft and unauthorized exposure. Users control their data. Encryption and access control.
Transparency	Communicates development process, capabilities, and limitations to stakeholders so they can make informed decisions.
Veracity & Robustness	System operates reliably even in unexpected situations, uncertainty, or errors. Resilient to input distribution changes.
Governance	Processes to define, implement, and enforce responsible AI practices within the organization.
Safety	Algorithms and models designed to avoid harm to humans or the environment. Proactively address bias and misuse.
Controllability	Ability to monitor and guide AI behavior to align with human values. Includes human oversight mechanisms.
Inclusivity	AI systems designed to be accessible and equitable across all demographic groups.

■ **EXAM TIP:** Transparency = HOW the model works (process/architecture). Explainability = WHY it made a specific decision. Interpretability = deep feature-weight level understanding.

Bias and Variance in Responsible AI

Problem	Description	Remediation
High Bias (Underfitting)	Model too simple — misses important data features. High gap between predicted and actual values. Performs poorly on TRAINING data.	Add features, increase model complexity, reduce regularization.
High Variance (Overfitting)	Model memorizes training noise — performs well on training but poorly on test data. Cannot generalize.	Add data, regularization, early stopping, simpler model, cross-validation.
Balanced (Ideal)	Low bias + low variance — generalizes well to unseen data.	Tune model complexity, data augmentation, proper train/test split.

Techniques to Overcome Bias/Variance

Technique	Description
Cross-validation	Train on subsets, evaluate on complementary subsets. Detects overfitting.
Increase training data	Add more samples to expand learning scope.
Regularization	Penalizes extreme weight values to prevent overfitting linear models.
Simpler models	Reduces overfitting by reducing model complexity.
Dimensionality reduction (PCA)	Reduces features while preserving information.
Early stopping	End training before model memorizes training data.
Data augmentation	Generate new instances of underrepresented groups to balance dataset.
Random oversampling/SMOTE	Balance imbalanced datasets for fair model training (SageMaker Data Wrangler).

Legal Risks of Working with GenAI

- Intellectual property infringement — verbatim reproduction of copyrighted training data.
- Biased model outputs causing discrimination or legal liability.
- Loss of customer trust from toxic, inaccurate, or inappropriate outputs.
- End-user risk from acting on hallucinated information.
- Plagiarism and cheating concerns (academic, professional contexts).
- Privacy violations — exposing PII from training data or user inputs.

Dataset Characteristics for Responsible AI

- **Inclusivity** — data collection processes are fair and unbiased.
- **Diversity** — data accurately represents all relevant groups and perspectives.
- **Balance** — adequate representation of each group; no dominant class skewing the model.
- **Curated data sources** — high-quality, relevant, clean, verified data.
- **Representativeness** — reflects the real-world distribution of the problem domain.
- **Governance** — documented data lineage, provenance, and collection methodology.

AWS Tools for Responsible AI ★

AWS Service/Tool	Responsible AI Function
Amazon SageMaker Clarify ★	Detect bias in datasets and models. Feature importance scores (which features most influence predictions). Tabular, NLP, computer vision support.
SageMaker Data Wrangler ★	Balance datasets: random undersampling, random oversampling, SMOTE. Low-code data prep.
SageMaker Model Monitor ★	Continuous or scheduled monitoring of production models. Detects data drift, performance degradation. Configurable alerts.
Amazon A2I ★	Augmented AI — human review workflows for low-confidence ML predictions.
Guardrails for Amazon Bedrock ★	Block undesirable topics, filter harmful content (hate/insults/sexual/violence), redact PII, enforce responsible AI policies.

Amazon Bedrock Model Evaluation ★	Automatic (accuracy, robustness, toxicity) and human evaluation (style, brand alignment). Compare FMs.
AWS AI Service Cards	Documentation for each AWS AI service: intended use, limitations, responsible AI design, deployment best practices.
Amazon SageMaker Model Cards ★	Catalog model details: intended use, risk rating, training info, evaluation results. Streamlined governance.
SageMaker Model Dashboard	Team view of model behavior in production. Centralized monitoring.
SageMaker Ground Truth	Human-in-the-loop data labeling + RLHF feedback (comparison and ranking data).
SageMaker Role Manager	Admins define minimum permissions in minutes for ML practitioners.

Task 4.2 — Transparent and Explainable Models

Transparency vs. Explainability vs. Interpretability ★

Concept	Meaning
Transparency	HOW the system works — development process, data sources, architecture communicated to stakeholders. System-level openness.
Explainability	WHY a specific decision was made — high-level, human-understandable justification of model behavior. Builds trust.
Interpretability	DEEP-LEVEL understanding — human can directly examine model weights and features to understand output. (e.g., linear regression coefficients).
Black Box Models	Lack transparency AND explainability. Complex neural networks with many layers. Hard to debug or audit.

■ **EXAM TIP:** Exam: 'Interpretability = deep internal mechanics understanding' vs. 'Explainability = high-level why a decision was made.' Both differ from Transparency (HOW the overall system works).

AWS Tools for Transparency & Explainability ★

AWS Tool	Role in Transparency/Explainability
Amazon SageMaker Model Cards ★	Transparency — document intended use, risk rating, training details, evaluation results. Governance and reporting. ★★
AWS AI Service Cards	Transparency — intended use cases, limitations, responsible AI design for each AWS AI service.
SageMaker Clarify ★	Explainability — feature importance scores showing WHICH features most influenced predictions. Bias detection.
SageMaker Autopilot ★	Explainability — per-instance explanations of predictions. Feature contribution scores during training and inference.

Safety vs. Transparency Trade-offs

Trade-off Area	Description
Accuracy	Complex (less interpretable) models often outperform simple (more interpretable) ones.
Privacy	Privacy-preserving techniques (differential privacy) improve safety but reduce model transparency.
Safety	Filtering/constraining model outputs for safety can reduce visibility into model reasoning.
Security	Air-gapped models (private networks) may be less open to external auditing.

RLHF — Reinforcement Learning from Human Feedback

- ML technique that uses human feedback to optimize models toward human-preferred behaviors.
- Humans provide ratings/preferences/demonstrations → reward model trained from ratings → RL refines the LLM.
- Used in both traditional AI and generative AI applications.
- **Amazon SageMaker Ground Truth** provides RLHF annotation capabilities (comparison and ranking data as reward function).
- **Benefits:** Enhances AI performance, supplies complex training parameters, increases user satisfaction.

Human-Centered Design (HCD) Principles for Explainable AI

HCD Principle	Description
Design for Amplified Decision-Making	Support decision-makers in high-stakes situations. Key aspects: Clarity, Simplicity, Usability, Reflexivity (prompt users to reflect), Accountability.
Design for Unbiased Decision-Making	Ensure processes are transparent, fair, and free of discriminatory biases. Equal opportunity for all stakeholders.
Design for Human & AI Learning	Create environments beneficial for both human and AI learning. Cognitive apprenticeship, personalization, user experience focus.

Task 5.1 — Securing AI Systems

AWS Shared Responsibility Model for AI ★

Party	Responsibility
AWS Responsibility	Security OF the cloud — physical infrastructure, hardware, networking, global infrastructure, hypervisor, managed service operations.
Customer Responsibility	Security IN the cloud — data, IAM configuration, application security, network controls, OS patching (for EC2), model access policies.

Defense in Depth for AI on AWS ★

Defense Layer	AWS Services & Approach
Data Protection	Encrypt at rest (AWS KMS, customer-managed keys). Encrypt in transit (TLS/SSL). S3 versioning for backup. AWS PrivateLink to keep data inside VPCs.
Identity & Access Mgmt	AWS IAM — users, roles, policies, least privilege. IAM Access Analyzer for overly permissive accounts. MFA.
Application Protection	AWS Shield (DDoS), Amazon Cognito (user authentication), WAF (web application firewall).
Network & Edge	Amazon VPC (isolated networks), Security Groups, Network ACLs, AWS WAF.
Infrastructure Protection	IAM user groups, Network ACLs, patching, vulnerability management.
Threat Detection	AWS Security Hub (single dashboard), Amazon GuardDuty (ML-based threat detection).
Incident Response	AWS Lambda + Amazon EventBridge for automated responses to security events.
Policies & Awareness	Least privilege via IAM Access Analyzer. Short-term credentials. Regular team training.

IN-SCOPE AWS Security Services ★

Service	AI Security Function
AWS IAM ★	Roles, policies, permissions controlling access to AI resources. Least privilege principle. ★★
Amazon Macie ★	ML-based service to discover and protect sensitive data (PII) in Amazon S3. Automated scanning.
AWS PrivateLink ★	Private connectivity to AWS services without traversing the public internet. Keep data in VPC.
AWS KMS ★	Key Management Service. Manage encryption keys for data at rest. Customer-managed keys.
AWS Secrets Manager ★	Store, rotate, and retrieve secrets (API keys, DB passwords). Reduce hard-coded credentials.

AI-Specific Security Considerations

Security Topic	Description
Prompt Injection	Malicious inputs override system instructions. Mitigate with input validation + Bedrock Guardrails.
Insecure Output Handling	Failure to sanitize model outputs leads to security vulnerabilities (XSS, command injection).
Encryption at Rest	KMS for training data, model artifacts, and outputs stored in S3 or other storage.
Encryption in Transit	TLS/SSL for all data movement between AI services.
Data Lineage	Track origin, transformation, and usage of data. Document provenance for compliance.
Data Cataloging	SageMaker Model Cards document data sources, model history, and metadata.
Application Security	Threat detection, vulnerability management, infrastructure protection for AI workloads.

Secure Data Engineering Best Practices

- Assess **data quality** before using for training — garbage in = garbage out.
- Implement **privacy-enhancing technologies** (differential privacy, data anonymization, pseudonymization).
- **Data access control** — IAM policies restricting who can read/write training data (least privilege).
- **Data integrity** — checksums, S3 versioning, pipeline validation.
- **Audit trails** — CloudTrail logs all data access and model operations.

Task 5.2 — Governance & Compliance for AI Systems

Regulatory Compliance Standards for AI ★

Standard	Description
ISO/IEC 27001	International standard for information security management. Controls for security best practices.
SOC (1/2/3)	AWS System and Organization Controls. Independent third-party assessment of AWS security controls.
NIST 800-53	US federal information system security controls. Formal evaluation for federal AI systems.
HIPAA	US healthcare data protection. AWS environment supports HIPAA-compliant AI workloads.
GDPR	EU data protection regulation. Strict requirements for personal data of EU citizens.
PCI DSS	Payment card industry security standard. For AI systems handling payment data.
Algorithm Accountability Laws	EU AI Act (transparency, risk assessment, human oversight). US state laws (NY Automated Decision Systems Law).

IN-SCOPE AWS Governance & Compliance Services ★

Service	Governance Function
AWS CloudTrail ★	Records ALL API calls and user activity. WHO did WHAT to WHICH resource WHEN. Operational and risk auditing. Essential for compliance. ★★

AWS Config ★	Tracks AWS resource configurations and relationships over time. Detects misconfigurations. Demonstrates compliance history. ★★
Amazon Inspector ★	Vulnerability management — continuously scans EC2, containers, Lambda for software vulnerabilities and network exposure.
AWS Audit Manager ★	Automates evidence collection for audits. Maps controls to frameworks (HIPAA, PCI, SOC). Manages stakeholder review.
AWS Artifact ★	On-demand downloads of AWS compliance docs (ISO certs, PCI reports, SOC reports). Share with auditors/regulators.
AWS Trusted Advisor ★	Best practice checks: cost, performance, security, resilience, operational excellence, service limits. Remediation recommendations. ★
Amazon CloudWatch	Monitoring metrics, logs, alarms. Real-time operational visibility.
AWS Well-Architected Tool	Reviews workloads against Well-Architected Framework pillars including Security and Operational Excellence.

Data Governance Strategies for AI

Concept	Description
Data Lifecycles	Manage data from creation → collection → processing → storage → consumption → disposal/archiving.
Data Logging	Systematic recording of inputs, outputs, model metrics, and system events. Enables debugging and monitoring.
Data Residency	Physical location of stored data. Critical for GDPR, data sovereignty, and proximity to compute.
Data Monitoring	Ongoing observation of data quality, anomalies, and data drift (input distribution changes over time).
Data Analysis/EDA	Statistical analysis, visualization, exploratory data analysis to understand data characteristics.
Data Retention	Policies defining how long data is kept. Balances regulatory requirements vs. retraining needs vs. cost.
Data Integrity	Accuracy, completeness, consistency. Data lineage and provenance tracking.
Data Privacy	Access controls, encryption, breach response, PII protection policies.

Governance Protocol Processes

Protocol Element	Description
Policies	Comprehensive policies: data management, model training, output validation, safety, human oversight, IP, bias, privacy. Regularly reviewed.
Review Cadence	Regular reviews (monthly/quarterly/bi-annually). Include technical, legal, and responsible AI perspectives with diverse stakeholders.
Review Strategies	Technical reviews (model performance, data quality) + non-technical reviews (policy alignment, regulatory compliance).

Governance Frameworks	Gen AI Security Scoping Matrix, Well-Architected Framework, NIST AI RMF, internal AI governance boards.
Transparency Standards	Publish model information, training data details, and key development decisions.
Team Training	Train all team members on bias mitigation, responsible AI practices, data governance best practices.

Appendix A — IN-SCOPE AWS Services (Full Official List) ★★

Machine Learning (Critical — Know All of These) ★★★

Service	Key Function
Amazon A2I	Human review workflows for ML predictions.
Amazon Bedrock	Managed FMs + RAG + agents + guardrails.
Amazon Comprehend	NLP: entities, key phrases, sentiment, PII.
Amazon Fraud Detector	Detect online fraud, account takeovers.
Amazon Kendra	Intelligent ML document search.
Amazon Lex	Build conversational chatbots (ASR + NLU).
Amazon Nova	Amazon's FM family on Bedrock.
Amazon Personalize	Real-time product/content recommendations.
Amazon Polly	Text-to-speech synthesis.
Amazon Q Developer	AI coding assistant.
Amazon Q Business	Enterprise AI assistant.
Amazon Rekognition	Image/video analysis: faces, objects, text.
Amazon SageMaker AI	End-to-end ML platform.
Amazon Textract	Extract text/data from scanned documents.
Amazon Transcribe	Automatic speech recognition (ASR).
Amazon Translate	Neural machine translation. 75+ languages.

Other IN-SCOPE Services (Supporting Categories)

Category	Services
Analytics	AWS Data Exchange, Amazon EMR, AWS Glue, AWS Glue DataBrew, AWS Lake Formation, Amazon OpenSearch Service ★, Amazon QuickSight, Amazon Redshift
Cloud Financial Mgmt	AWS Budgets, AWS Cost Explorer
Compute	Amazon EC2
Containers	Amazon ECS, Amazon EKS
Database ★	Amazon DocumentDB, Amazon DynamoDB, Amazon ElastiCache, Amazon MemoryDB, Amazon Neptune ★, Amazon RDS ★
Management & Governance ★	AWS CloudTrail ★, Amazon CloudWatch, AWS Config ★, AWS Trusted Advisor ★, AWS Well-Architected Tool
Networking	Amazon CloudFront, Amazon VPC

Security, Identity & Compliance ★	AWS Artifact, AWS Audit Manager, AWS IAM ★, Amazon Inspector, AWS KMS ★, Amazon Macie ★, AWS Secrets Manager ★
Storage	Amazon S3 ★, Amazon S3 Glacier

Appendix B — OUT-OF-SCOPE AWS Services (Lower Priority)

■ **IMPORTANT:** These services appear on the exam but are out-of-scope — you should recognize them but NOT be tested on their details.

Category	Out-of-Scope Services
Analytics OOS	AWS Clean Rooms, Amazon CloudSearch, Amazon FinSpace, Amazon MSK
Application Integration OOS	Amazon AppFlow, Amazon MQ, Amazon SWF
Business Applications OOS	Amazon Chime, Amazon Honeycode, Amazon Pinpoint, Amazon SES, AWS Supply Chain, AWS Wickr, Amazon WorkDocs, Amazon WorkMail
Compute OOS	AWS App Runner, AWS Elastic Beanstalk, EC2 Image Builder, Amazon Lightsail
Database OOS	Amazon Keyspaces (Cassandra), Amazon QLDB, Amazon Timestream
ML OOS (surprising!)	AWS DeepComposer, AWS HealthImaging, AWS HealthOmics, Amazon Monitron, AWS Panorama
Mgmt & Gov OOS	AWS Control Tower, AWS Organizations, AWS OpsWorks, AWS Proton, AWS Resilience Hub, AWS Systems Manager Incident Manager
Security OOS	AWS Certificate Manager, AWS CloudHSM, Amazon Cognito, Amazon Detective, AWS Directory Service, AWS Firewall Manager, Amazon GuardDuty, AWS IAM Identity Center, AWS Security Hub, Amazon Security Lake, AWS Shield, AWS WAF
IoT OOS (all)	AWS IoT Analytics, AWS IoT Core, AWS IoT Device Defender, AWS IoT Device Management, AWS IoT Events, FreeRTOS, AWS IoT Greengrass, AWS IoT SiteWise
Media OOS (all)	Amazon Elastic Transcoder, AWS Elemental MediaConvert/Live/Package/Store/Tailor, Amazon IVS, Amazon Nimble Studio
Migration OOS (all)	AWS DMS, AWS DataSync, AWS Migration Hub, AWS Snow Family, AWS Transfer Family, AWS Application Migration Service
Networking OOS	AWS App Mesh, AWS Direct Connect, AWS Global Accelerator, AWS Private 5G, Amazon Route 53, Amazon VPC IPAM
Storage OOS	AWS Backup, AWS Elastic Disaster Recovery

Quick Reference — Critical Exam Distinctions ★★

Concept A		Concept B	Key Distinction
Fine-tuning	v s .	RAG	Fine-tuning CHANGES model weights. RAG retrieves at inference time — NO weight changes.
Transfer Learning	v s .	Domain Adaptation	Transfer Learning re-uses model from different task. Domain Adaptation fine-tunes with limited domain-specific data.
Transparency	v s .	Explainability	Transparency = HOW the system works. Explainability = WHY a specific decision was made.
Explainability	v s .	Interpretability	Explainability = high-level why. Interpretability = deep weight/feature-level understanding.
ROUGE (Recall)	v s .	BLEU (Precision)	ROUGE measures recall (key info captured) — summarization. BLEU measures precision — translation.
Batch Inference	v s .	Real-time Inference	Batch = offline, large datasets, cost-effective. Real-time = low latency, persistent endpoint, immediate.
Underfitting	v s .	Overfitting	Underfitting = High Bias (bad on train). Overfitting = High Variance (good on train, bad on test).
AWS Responsibility	v s .	Customer Responsibility	AWS = security OF the cloud. Customer = security IN the cloud.
Foundation Model	v s .	Traditional ML model	FM = broad, multi-task, one model. Traditional ML = narrow, single-task, separate model per task.
SageMaker Clarify	v s .	SageMaker Model Cards	Clarify = bias detection + feature importance. Model Cards = documentation/transparency governance.
CloudTrail	v s .	Config	CloudTrail = WHO did WHAT WHEN (API logs). Config = HOW resources are configured over time.
Temperature low (0)	v s .	Temperature high (1+)	Low = deterministic/focused outputs. High = creative/random outputs.
Amazon Macie	v s .	Amazon Inspector	Macie = finds PII/sensitive data in S3. Inspector = scans for software vulnerabilities in compute.

GenAI Application Lifecycle — 5 Stages ★

Stage	Key Activities
1. Define Use Case	Analyze app needs, user requirements, business goals. Define KPIs. Align stakeholders.
2. Select FM	Choose from pre-trained FMs or build from scratch. Consider cost, modality, latency, compliance.
3. Improve Performance	Prompt engineering (fastest/cheapest) → RAG → Fine-tuning (most expensive). Iterate.
4. Evaluate Results	ROUGE, BLEU, BERTScore. Human evaluation. Benchmark datasets. Test edge cases.
5. Deploy Application	Production deployment. Monitor cost, regions, quotas, security (shared responsibility).

Top 15 Exam Tips

- **1. Domain 3 = 28%:** Spend the most study time on Applications of Foundation Models — RAG, prompt engineering, fine-tuning, evaluation metrics.
- **2. Amazon Bedrock:** Unified API, serverless, Knowledge Bases (RAG), Agents, Guardrails, Model Evaluation — know all features.
- **3. ROUGE vs BLEU vs BERTScore:** ROUGE = recall/summarization. BLEU = precision/translation. BERTScore = semantic similarity.
- **4. RAG vs Fine-tuning:** RAG = no weight changes, retrieves at inference. Fine-tuning = changes model weights, needs labeled data.
- **5. Customization Ladder:** Use existing → Prompt Eng → RAG → Fine-tune → Continuous Pre-train → Build from scratch (cost increases).
- **6. Few-shot prompting:** Provide 2–5 examples in the prompt. LEAST operational effort for style/format matching.
- **7. CloudTrail vs Config:** CloudTrail = API activity log (who/what/when). Config = resource configuration history (how it's configured).
- **8. SageMaker Clarify:** Bias detection + feature importance. Integrated with SageMaker Experiments.
- **9. Temperature parameter:** Low = deterministic/focused. High = creative/varied. Controls output randomness.
- **10. Out-of-scope tasks:** NOT tested: coding/building models, feature engineering math, hyperparameter tuning math, deploying ML pipelines.
- **11. Shared responsibility:** AWS = security OF the cloud (infra). Customer = security IN the cloud (data, config, IAM).
- **12. Model Cards:** SageMaker Model Cards = transparency/governance documentation. NOT for bias detection (that's Clarify).
- **13. Overfitting vs Underfitting:** Overfitting = high variance (good train, bad test). Underfitting = high bias (bad both). Goal: low bias + low variance.
- **14. Pass score:** 700/1,000 scaled. 50 scored questions + 15 unscored (not identified). No penalty for guessing.
- **15. Amazon Inspector vs Macie:** Inspector = software vulnerability scanning. Macie = sensitive data/PII discovery in S3.

✓ **KEY FACT:** You only need to pass the OVERALL exam (700/1000). Compensatory scoring — no minimum per domain. Focus extra time on Domain 3 (28%) and Domain 2 (24%).